

Course Plan

Semester: V

Year: 2025-26

Course Title: Natural Language Processing with GenAI	Course Code: 24EBCC304
Total Contact Hours: 30+48	Duration of ESA: 2 Hours
ISA Marks: 50	ESA Marks: 50
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Checked By: Dr. S. G. Totad	Date: 12/07/2025

Prerequisites:

The subject requires the students to know about Data Mining, Machine Learning, and Python Programming.

Course Outcomes (COs):

At the end of the course, the student should be able to:

1. Explain the fundamentals of Natural Language Processing (NLP) and apply essential text preprocessing techniques for language understanding.
2. Explain parsing techniques and language models, including Transformers, to implement machine translation and text generation using attention-based architectures.
3. Apply Generative AI techniques and leverage Large Language Models (LLMs) to build practical solutions for real-world natural language processing applications.

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Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes

Course Title: Natural Language Processing with GenAI	Semester: V
Course Code: 24EBCC304	Year: 2025-26

Course Outcomes (COs) / Program Outcomes (POs)	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Explain the fundamentals of Natural Language Processing (NLP) and apply essential text preprocessing techniques for language understanding.		M	M		M									
2. Explain parsing techniques and language models, including Transformers, to implement machine translation and text generation using attention-based architectures.		M	M		M									
3. Apply Generative AI techniques and leverage Large Language Models (LLMs) to build practical solutions for real-world natural language processing applications.		H	H		H	M	M			M				

Degree of compliance L: Low M: Medium H: High

Competency addressed in the Course and corresponding Performance Indicators

Competency: 2.1	Demonstrate an ability to identify and characterize a computing problem characterize a computing problem
PI Code :2.1.2	Identify required input, output, mathematical model and logic to solve computing problems.
Competency: 2.2	Demonstrate an ability to formulate a solution plan and methodology for a computing problem.
PI Code: 2.2.3	Identify solution methods for solving the problem along with justified approximations and assumptions.
Competency: 2.2	Demonstrate an ability to formulate a solution plan and methodology for a computing problem.
PI Code: 2.2.4	Compare and contrast alternative solution plans to select the best plan.
Competency: 2.3	Demonstrate an ability to formulate and interpret a model.
PI Code: 2.3.2	Identify assumptions (mathematical and physical) necessary to allow modeling of a system at the level of accuracy required.
Competency: 3.1	Demonstrate an ability to define a complex open-ended problem in computing terms.
PI Code: 3.1.4	Extract software engineering requirements that meet appropriate consideration for public health and safety, cultural, societal, and environmental considerations as per the Codes and Standards.
Competency: 3.2	Demonstrate an ability to generate a diverse set of alternative design solutions.
PI Code: 3.2.2	Build models, prototypes, etc., to develop diverse set of design solutions.
Competency: 3.3	Demonstrate an ability to select the optimal design scheme for further development.
PI Code: 3.3.2	Consult with domain experts and stakeholders to select proposed computing design solution for further development.
Competency: 5.2	Demonstrate an ability to select and apply domain specific tools, techniques and resources.
PI Code: 5.2.2	Demonstrate proficiency in using domain specific tools.
Competency: 6.1	Demonstrate an ability to recognize ethical issues.
PI Code: 6.1.1	Identify situations of unethical professional conduct and propose ethical alternatives.
Competency: 7.2	Demonstrate an ability to Identify changing trends in computing knowledge and practice
PI Code: 7.2.1	Identify technological advances in computing that required practitioners to stay updated with current technologies

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Competency10.1	Demonstrate the ability to describe computing roles in a broader context, e.g. as pertains to the environment, health, safety, and public welfare.
PI Code: 10.1.2	Able to demonstrate standards, best practices and code of ethics in the application development favorable to the public interest and public protection.

Eg: 1.2.3: Represents Program Outcome '1', Competency '2' and Performance Indicators '3'.

Chapter-wise CO Mapping

	Ch.1	Ch.2	Ch.3	Ch.4	Ch.5	Ch.6
CO1	√					
CO2		√	√			
CO3				√	√	√

Chapter 1: Introduction to NLP and Preprocessing

Topics/Concepts: NLP overview & applications, Challenges in language understanding, Basic components of NLP systems, Text preprocessing: Tokenization, Stop word removal, Stemming, Lemmatization, POS tagging, Bag of Words, Introduction to word embeddings.

Course Outcome 1: Explain the fundamentals of Natural Language Processing (NLP) and apply essential text preprocessing techniques for language understanding.

Competency	Performance Indicators (PIs)	Breaking Down at Concept Level	Course Specific performance indicators (CSPIs)
2.1 Demonstrate an ability to identify and characterize a computing problem	2.1.2: Identify required input, output, mathematical model, and logic to solve computing problems	- Define NLP and its applications in real-world systems. - Identify real world examples of NLP applications - Recognize challenges in language understanding.	Relate foundational concepts and challenges of NLP to practical applications.
2.2 Demonstrate an ability to formulate a solution plan and methodology for a computing problem.	2.2.3: Identify solution methods with justified approximations and assumptions.	- Apply appropriate preprocessing techniques. - Justify the use of specific preprocessing steps based on the problem domain.	Select and justify preprocessing steps required for a given NLP application.
5.2 Demonstrate an ability to select and apply domain-specific tools, techniques, and resources	5.2.2 Demonstrate proficiency in using domain-specific tools.	Use NLTK and SpaCy libraries to implement preprocessing steps.	Apply text preprocessing and tagging using NLP libraries on real-world datasets.

Chapter 2: Parsing and Syntax Analysis

Topics/Concepts: Grammar rules in NLP, Dependency and Constituency Parsing, Parsing techniques and algorithms, Introduction to neural dependency parsing.

Course Outcome 2: Explain parsing techniques and language models, including Transformers, to implement machine translation and text generation using attention-based architectures.

Competency	Performance Indicators (PIs)	Breaking Down At Concept Level	Course Specific performance indicators (CSPIs)
2.1 Demonstrate ability to identify and characterize a computing problem	2.1.2 Identify required input, output, mathematical model, and logic.	- Explain syntactic structures in natural language - Analyze the structure and role of grammar rules, such as Context-Free Grammars (CFGs), in syntactic parsing.	Evaluate grammatical composition and formulate the parsing workflow in NLP applications.
2.2 Demonstrate ability to formulate a solution plan and methodology	2.2.3 Identify solution methods with justified approximations and assumptions.	- Apply parsing techniques like top-down, bottom-up, shift-reduce parsing - Analyze underlying assumptions such as non-ambiguity and left-most derivation in parsing strategies.	Determine the most effective parsing technique for handling diverse grammatical structures and sentence forms.
2.2 Demonstrate ability to formulate a solution plan and methodology	2.2.4 Compare and contrast alternative solution plans	- Analyze the differences between dependency parsing and constituency parsing in representing sentence structure. - Assess the strengths and limitations of neural parsing approaches in comparison to traditional rule-based methods.	Justify selection of parsing model based on performance and language coverage.

Chapter 3: Language Models & Transformers

Topics/concepts: N-grams and statistical models, Neural language models: Word2Vec, Glove, Transformers for text understanding and generation, Sequence-to-sequence models, Attention mechanisms

Course Outcome 2: Explain parsing techniques and language models, including Transformers, to implement machine translation and text generation using attention-based architectures.

Competency	Performance Indicators (PIs)	Breaking Down at Concept Level	Course Specific performance indicators (CSPIs)
2.1 Identify and characterize a computing problem	2.1.2 Identify required input, output, mathematical model, and logic	- Analyze the task of language modeling as the prediction of upcoming tokens within a given text context. - Identify training data as the input and interpret the model's output as either probability distributions over vocabulary or generated text sequences. - Identify both statistical (n-gram) and neural models as viable solutions for sequence prediction tasks.	Explain the functions of language modeling and compare how n-gram and neural models handle input-output flows.
2.2 Formulate a solution plan and methodology	2.2.3 Identify solution methods with justified assumptions	- Apply n-gram models and note assumptions. - Apply neural language models: Word2Vec, GloVe, RNNs, Transformers	Justify the selection of either statistical or neural language models based on the nature of the task and the complexity of the language involved.
2.2 Formulate a solution plan and methodology	2.2.4 Compare and contrast alternative solution plans	- Compare n-gram vs Word2Vec vs Transformer models - Explain trade-offs: interpretability, training time, accuracy, context handling	Compare effectiveness and performance of classical and modern language models for generation and understanding
3.2 Generate diverse design solutions	3.2.2 Build models, prototypes, or solutions	- Implement Word2Vec, Transformer encoder-decoder models - Build sequence-to-sequence models for translation or summarization.	Design and develop working prototypes for text generation or translation using modern models
3.3 Select optimal design schemes	3.3.2 Consult and choose the best computing design	Evaluate different architectures based on accuracy, data availability, and compute requirements	Choose the most effective language model for a given application scenario by comparing alternatives
5.2 Use domain-specific tools	5.2.2 Demonstrate proficiency with NLP tools	- Use TensorFlow, PyTorch, HuggingFace Transformers to train/evaluate models - Experiment with pre-trained models	Apply domain tools to build and fine-tune neural language models for real-world tasks

Chapter 4: Machine Translation & Generation

Topics/concepts: Overview of Machine Translation, Statistical vs Neural Machine Translation, Seq2Seq with attention, Text generation techniques: LSTM, Transformers, Role of Auto encoders & Decoders

Course Outcome 2: Explain parsing techniques and language models, including Transformers, to implement machine translation and text generation using attention-based architectures.

Competency	Performance Indicators (PIs)	Breaking Down at Concept Level	Course Specific performance indicators (CSPIs)
2.1 Identify and characterize a computing problem	2.1.2 Identify input, output, model, and logic	Explain machine translation as the process of generating equivalent text in a target language from a source language input.	Describe how machine translation architectures process input text to generate translated output through an end-to-end pipeline.
2.2 Formulate a solution plan and methodology	2.2.3 Identify solution methods with justified assumptions	Analyze the structure of Seq2Seq models that utilize encoder-decoder mechanisms based on LSTM or Transformer architectures.	Identify and apply the most effective translation or generation approach based on task-specific characteristics.
2.2 Formulate a solution plan and methodology	2.2.4 Compare and contrast solution plans	- Analyze the key characteristics and performance differences between Statistical and Neural Machine Translation approaches. - Analyze the differences between LSTM and Transformer architectures in the context of text generation. - Compare decoding methods such as greedy and beam search with respect to their effectiveness in text generation tasks.	Select the most suitable translation or generation model by evaluating its performance and adaptability to task requirements.
3.3 Select optimal design scheme	3.3.2 Consult with domain experts to select design	- Determine the most suitable approach among SMT, LSTM, and Transformer models by considering task requirements and data constraints. - Analyze the trade-offs involved in achieving high performance versus maintaining model simplicity.	Identify and choose the most appropriate model for performing machine translation or text generation based on task requirements.
5.2 Use domain-specific tools	5.2.2 Demonstrate proficiency in using tools	Use libraries like TensorFlow, PyTorch, or Hugging Face for translation and generation	Develop machine translation and text generation models using widely adopted deep learning frameworks.

Chapter 5: Generative AI and Advanced Architectures

Topics/Concepts: Discriminative vs Generative models, Generative Adversarial Networks (GANs) in NLP, Types of GANs, Diffusion models: concepts & types, Introduction to Chatbot and Dialogue Systems

Course Outcomes 3: Apply Generative AI techniques and leverage Large Language Models (LLMs) to build practical solutions for real-world natural language processing applications.

Competency	Performance Indicators (PIs)	Breaking Down At Concept Level	Course Specific performance indicators (CSPIs)
2.2 Formulate a solution plan and methodology	2.2.3 Identify solution methods with justified assumptions	- Apply GAN architectures to solve natural language processing problems involving text generation. - Analyze the fundamental concepts and classifications of diffusion models in the context of text generation	Apply GANs and diffusion models for text generation in NLP, and analyze their core concepts and types.
2.2 Formulate a solution plan and methodology	2.2.4 Compare and contrast solution plans	- Compare the modeling approaches and use cases of discriminative and generative models in NLP. - Evaluate the strengths and limitations of GANs and diffusion models for realistic text or image generation.	Determine the most suitable generative method based on task-specific output goals.
3.2 Generate diverse design solutions	3.2.2 Build models/prototypes, etc.	- Implement basic GAN architectures for generating text or structured data in NLP applications. - Create chatbot prototypes leveraging predefined dialogue templates or transformer encoders for natural language understanding.	Build and test generative model prototypes tailored to tasks like dialogue generation and content creation.
3.3 Select optimal design scheme	3.3.2 Consult with domain experts to select design	Choose between GANs, diffusion models, or transformer-based dialogue systems based on complexity and data availability	Determine the best generative architecture aligned with real-world NLP use cases like summarization or dialogue generation.
5.2 Use domain-specific tools	5.2.2 Demonstrate proficiency using domain-specific tools	Implement and fine-tune generative models using tools like PyTorch, HuggingFace, OpenAI API.	Build and assess NLP generation models using widely used tools and frameworks.

Chapter 6: Large Language Models (LLMs)

Topics/Concepts: Introduction to LLMs: GPT, BERT, Prompting techniques for GenAI, Adapters and LoRA (Low-Rank Adaptation), Applications and limitations of LLMs

Course Outcomes 3: Apply Generative AI techniques and leverage Large Language Models (LLMs) to build practical solutions for real-world natural language processing applications.

Competency	Performance Indicators (PIs)	Breaking Down At Concept Level	Course Specific performance indicators (CSPIs)
2.1 Identify and characterize a computing problem	2.1.2 Identify input, output, model, and logic	- Explain the architecture and functioning of LLMs in generating and interpreting natural language - Identify the input-output structure of LLMs, where prompts initiate text generation and responses are the resulting outputs - Explain the underlying architecture and processing models like GPT and BERT.	Describe the mechanism by which LLMs use pretrained knowledge to interpret prompts and generate coherent outputs.
2.2 Formulate a solution plan and methodology	2.2.3 Identify solution methods with justified assumptions	- Apply prompt-based learning strategies to enable models to generalize from few examples. - Use parameter-efficient approaches like adapters or LoRA to fine-tune language models without retraining the entire architecture. - Recognize key assumptions behind pretraining and adapting models to new tasks.	Choose and apply effective prompt-based or model adaptation approaches tailored to different NLP tasks.
2.2 Formulate a solution plan and methodology	2.2.4 Compare and contrast solution plans	- Compare full fine-tuning, adapter tuning, and LoRA for efficiency and flexibility. - Evaluate prompting versus full model retraining for task effectiveness. - Discuss trade-offs in resource usage, accuracy, and suitability for different use cases	Identify the optimal way to apply LLMs depending on the use case and resource availability.
2.3 Formulate and interpret a model	2.3.2 Identify modeling assumptions	Comprehend the underlying assumptions involved in pre-training, such as masked language modeling used in BERT.	Identify assumptions and limitations impacting LLM behavior and generalization.

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		Identify the constraints of large language models related to context window size, memory capacity, and inherent biases.	
3.2 Generate diverse design solutions	3.2.2 Build models/prototypes, etc.	- Develop small-scale applications utilizing LLM APIs like ChatGPT or HuggingFace. - Create prototypes of Generative AI applications using prompting.	Build basic Generative AI applications, such as summarizers and question-answering systems, using large language models (LLMs).
3.3 Select optimal design scheme	3.3.2 Consult to select design	- Select the appropriate deployment method among fine-tuning, adapter-based tuning, and prompt engineering. - Assess LLM designs for applications in areas such as education, healthcare, and content creation	Choose the optimal LLM approach for real-world applications by balancing performance and cost.
5.2 Use domain-specific tools	5.2.2 Demonstrate proficiency in tools	- Utilize LLM APIs such as OpenAI, HuggingFace, or others. - Apply LoRA tuning using libraries like PEFT or Transformers	Apply LLM toolkits and APIs for building GenAI features
7.2 Identify changing trends in computing	7.2.1 Identify technological advances	- Examine the evolution and expansion of GPT-style models. - Explain the differences and trends between open-source and proprietary large language models. - Recognize emerging challenges in the development and deployment of LLMs	Track the rapid evolution of LLM technologies and their impact on NLP applications

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Program: Bachelor of Computer Applications		Semester: V
Course Title: Natural Language Processing with Gen AI		Course Code:24EBCC304
L-T-P: 2-0-2	Credits: 4	Contact Hrs: 6 hrs/week
ISA Marks: 50	ESA Marks: 50	Total Marks: 100
Teaching Hrs: 30	Practical Hours: 48	Exam Duration:2 Hrs
Unit I		
1	Introduction to NLP & Preprocessing NLP overview & applications, Challenges in language understanding, Basic components of NLP systems, Text preprocessing: Tokenization, Stop word removal, Stemming, Lemmatization, POS tagging, Bag of Words, Introduction to word embedding's.	5 hrs
2	Parsing and Syntax Analysis Grammar rules in NLP, Dependency and Constituency Parsing, Parsing techniques and algorithms, Introduction to neural dependency parsing.	4 hrs
3	Language Models & Transformers N-grams and statistical models, Neural language models: Word2Vec, Glove, Transformers for text understanding and generation, Sequence-to-sequence models, Attention mechanisms.	6 hrs
Unit II		
4	Machine Translation & Generation Overview of Machine Translation, Statistical vs Neural Machine Translation, Seq2Seq with attention, Text generation techniques: LSTM, Transformers, Role of Auto encoders & Decoders.	5 hrs
5	Generative AI and Advanced Architectures Discriminative vs Generative models, Generative Adversarial Networks (GANs) in NLP, Types of GANs, Diffusion models: concepts & types, Introduction to Chabot and Dialogue Systems.	5 hrs
6	Large Language Models (LLMs) Introduction to LLMs: GPT, BERT, Prompting techniques for GenAI, Adapters and LoRA (Low-Rank Adaptation), Applications and limitations of LLMs.	5 hrs
Text Books: 1. Practical Natural Language Processing: A Comprehensive Guide to Building Real World NLP Systems-Sowmya Vajjala,Bodhisattwa Majumder,Anuj Gupta and Harshit Surana,2022 2. Hands on Generative AI with Transformers and Diffusion Models-Omar Sanseviero, Pedro Cuenca,Apolinario Passos,Jonathan Whitkar,2024		
Reference Books: 1. Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play by David Foster.2019		

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Evaluation Scheme

ISA Assessment	ISA MARKS	ESA Assessment	ESA MARKS
ISA- 1	10	ESA THEORY	25
ISA- 2	10		
Assignment	05		
ISA Lab Practices and Lab Activities	25	ESA LAB	25

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Expr. No.	Integrated Practices	Sessions
1	Write a Python program to parse HTML content and perform advanced text preprocessing operations such as tokenization, stop word removal, stemming, and lemmatization.	1
2	Implement a Python program to train and load Word2Vec (or GloVe) embeddings on a sample corpus and display the similar words using cosine similarity.	2
3	Write a Python program to extract and display all unigrams, bigrams from the given text corpus.	1
4	Implement a Python program to generate the TF-IDF matrix for a set of documents and use it for document similarity comparison or text classification.	1
5	Implement a Python program to build a simple Sequence-to-Sequence model using an encoder-decoder architecture for the task English to any other language.	2
6	Write a Python program to implement an LSTM model for text generation or sentiment analysis on a sample dataset.	2
7	Implement a Python program to perform text classification using pre-trained BERT and GPT models via the Hugging Face Transformers library.	1
8	Develop a course-project using NLP and Generative AI techniques to solve a real-world problem.	2

Rubric for Integrated Practices

Criteria	Excellent	Good	Satisfactory	Needs Improvement
Problem Understanding (20M)	Demonstrates an exceptional understanding of the problem requirements and objectives.	Shows a good understanding of the problem requirements and objectives.	Demonstrates a basic understanding of the problem requirements and objectives.	Demonstrates a limited understanding of the problem requirements and objectives.
Program Execution (20M)	Program is executed with a high level of skill, precision, and creativity.	Program is well-executed with a good level of skill and attention to detail.	Program is adequately executed with some room for improvement.	Program execution lacks skill and attention to detail.
Problem Solving (20M)	Demonstrates exceptional problem-solving skills throughout the lab.	Effectively identifies and resolves issues as they arise.	Adequately addresses problems but may struggle with more complex issues.	Struggles to identify and resolve problems, impacting lab progress.
Presentation Skills (10M)	Exceptional presentation skills, effectively communicates the lab's key aspects.	Good presentation skills, presenting key aspects of the lab clearly.	Adequate presentation skills but may lack clarity or engagement.	Presentation lacks structure and fails to effectively convey lab details.
Creativity and Innovation (10M)	Demonstrates outstanding creativity and innovation in lab design and implementation.	Shows creativity and innovation in lab design and implementation.	Demonstrates some creativity, but the lab lacks innovation.	Lacks creativity and innovation, resulting in a conventional lab.
Adherence to Guidelines (10M)	Strict adherence to lab guidelines and requirements.	Generally, adheres to lab guidelines with minor deviations.	Some deviations from lab guidelines, impacting overall quality.	Frequent deviations from lab guidelines, significantly impacting quality.
Overall Contribution to Course (10M)	Exceptional contribution to the course through the lab's depth and relevance.	Good contribution to the course through a well-executed lab.	Adequate contribution to the course, but the lab's impact could be greater.	Limited contribution to the course, with minimal impact from the lab.

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Course Unitization for Minor exams and End Semester Assessment

Topics / Chapters	Teaching hours	No. of Questions in ISA-1	No. of Questions in ISA-2	No. of Questions in ESA EXAM
Unit I				
1. Introduction to NLP and Preprocessing	5	1	--	1
2. Parsing and Syntax Analysis	4	1	--	1
3. Language Models and Transformers	6	1	--	1
Unit II				
4. Machine Translation and Generation	5	--	1	1
5. Generative AI and advanced Architectures	5	--	1	1
6. Large Language Models (LLMs)	5	--	1	1

Note

- Unit I and II will have three full questions of 15 marks each of which any two to be solved from each unit.
- Questions in Unit I and II, the questions from different chapters can be mixed
- All full questions may have maximum of 3 sub questions depending on the course requirement

Date:

Head of Department

Course Assessment Plan

Course Title: Natural Language Processing with GenAI Code:24EBCC304						
Course outcomes (COs)	Weightage in assessment	Assessment Methods				
		ISA-1	ISA-2	Seminar	Lab	Semester end exam
1. Explain the fundamentals of Natural Language Processing (NLP) and apply essential text preprocessing techniques for language understanding.	27%	✓			✓	✓
2. Explain parsing techniques and language models, including Transformers, to implement machine translation and text generation using attention-based architectures.	36%	✓	✓		✓	✓
3. Apply Generative AI techniques and leverage Large Language Models (LLMs) to build practical solutions for real-world natural language processing applications.	36%		✓	✓	✓	✓
Weightage	100%	10%	15%	05%	20%	50%

Chapter wise Plan

Course Code and Title: 24EBCC304/Natural Language Processing with GenAI	
Chapter Number and Title: Chapter 1: Introduction to NLP and Preprocessing	Planned Hours: 5hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain the fundamentals, applications, and challenges of Natural Language Processing.	CO1	L2	2.1
2. Apply essential text preprocessing techniques.	CO1	L2	2.2
3. Construct text representations using Bag of Words and explain the concept of word embeddings.	CO1	L3	2.2

Lesson Schedule

Class No. - Portion covered per hour / per Class

1. NLP overview & applications.
2. Challenges in language understanding and Basic components of NLP systems.
3. Text preprocessing: Tokenization, Stop word removal.
4. Stemming, Lemmatization, POS tagging.
5. Bag of Words, Introduction to word embedding's.

Review Questions

Questions	TLOs	BL	PI Code
1. Define NLP. Explain the key applications of Natural Language Processing (NLP) in real-world domains.	TLO1	L2	2.1.2
2. List and Explain Challenges of Natural Language Processing.	TLO1	L2	2.1.2
3. Explain the Basic Components of NLP Systems.	TLO1	L2	2.1.2
4. Explain the process of text preprocessing in Natural Language Processing (NLP) with the help of the following sentence: <i>"The students are studying natural language processing in the lab."</i> Describe the steps involved such as tokenization, stop word removal, stemming, and lemmatization.	TLO2	L3	2.2.3
5. What is the Bag of Words model? How is it constructed and used in NLP tasks? Explain with an example.	TLO3	L2	2.3.2

Course Code and Title: 24EBCC304/Natural Language Processing with GenAI		
Chapter Number and Title: Chapter 2: Parsing and Syntax Analysis		Planned Hours: 04hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain grammar rules, and distinguish between dependency and constituency parsing techniques in NLP.	CO2	L2	2.1
2. Apply standard parsing algorithms to analyze sentence structures using dependency and constituency grammar.	CO2	L3	2.2
3. Interpret the concept of neural dependency parsing and compare it with rule-based parsing methods.	CO2	L4	2.2

Lesson Schedule

Class No. - Portion covered per hour / per Class

- Grammar rules in NLP
- Dependency and Constituency Parsing
- Parsing techniques and algorithms
- Introduction to neural dependency parsing

Review Questions

Sl.No. - Questions	TLOs	BL	PI Code
1. Explain the role of grammar rules in Natural Language Processing.	TLO1	L2	2.1.2
2. Explain various parsing algorithms used in NLP. Compare top-down and bottom-up parsing with examples.	TLO3	L3	2.2.3
3. What is neural dependency parsing? Explain its architecture and how it improves traditional parsing methods.	TLO2	L2	2.2.4
4. Draw a basic dependency tree for the sentence "John eats an apple." Explain the dependency relations.	TLO2	L3	2.2.3
5. Write short notes on: (a) Shift-reduce parsing (b) CYK algorithm	TLO2	L2	2.2.3

Course Code and Title: 24EBCC304/Natural Language Processing with GenAI		
Chapter Number and Title: Chapter 3: Language Models & Transformers		Planned Hours: 06Hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain statistical language modeling using N-grams and understand the limitations leading to neural approaches.	CO2	L2	2.1
2. Apply and compare neural language models such as Word2Vec and GloVe for text representation	CO2	L4	2.2
3. Illustrate the architecture and functioning of Transformers, including attention mechanisms and sequence-to-sequence models.	CO2	L3	2.2

Lesson Schedule
Class No. - Portion covered per hour / per Class
1. N-grams and statistical models
2. Neural language models: Word2Vec
3. Neural language models: GloVe
4. Transformers for text understanding and generation
5. Sequence-to-sequence models
6. Attention mechanisms

Review Questions

Sl.No. - Questions	TLOs	BL	PI Code
1. What is n-grams? Explain unigram, bigram, and trigram models.	TLO1	L2	2.1.2
2. What is Word2Vec? Explain the difference between CBOW and Skip-gram models.	TLO2	L3	2.2.3
3. What is GloVe? Describe the training objective and how it captures word relationships using co-occurrence matrices.	TLO2	L3	2.2.3
4. Explain Sequence-to-Sequence models. How does the attention mechanism enhance the performance of Seq2Seq models?	TLO3	L2	2.2.4

Course Code and Title: 24EBCC304 / Natural Language Processing with GenAI
Chapter Number and Title: Chapter 4: Machine Translation & Generation
Planned Hours:05 Hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

TLO's	CO's	BL	CA Code
1. Explain statistical and neural approaches to machine translation.	CO2	L2	2.1
2. Illustrate the architecture and working of Seq2Seq models with attention for text translation.	CO2	L3	2.2
3. Compare LSTM, Transformers, and autoencoders in text generation tasks.	CO2	L4	2.2

Lesson Schedule

Class No. - Portion covered per hour

1. Overview of Machine Translation
2. Statistical vs Neural Machine Translation
3. Text generation techniques: LSTM
4. Transformers
5. Role of Autoencoders & Decoders

Review Questions

Sl.NO. Questions	TLO	BL	PI Code
1. What is Machine Translation? Explain its types with examples	TLO1	L2	2.1.2
2. Compare Statistical Machine Translation (SMT) and Neural Machine Translation (NMT) in terms of architecture and performance.	TLO1	L3	2.2.4
3. Explain how LSTM is used in text generation tasks with examples	TLO3	L2	3.3.2
4. Describe the Transformer architecture and its role in text understanding and generation.	TLO3	L3	3.3.2
5. What is the role of auto encoders and decoders in NLP? Explain their structure and use in text reconstruction or representation.	TLO3	L3	2.24

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Course Code and Title: 24EBCC304/Natural Language Processing with GenAI	
Chapter Number and Title: Chapter 5: Generative AI & Advanced Architectures	Planned Hours: 05 Hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

TLO's	CO's	BL	CA Code
1. Differentiate between discriminative and generative models with example relevant to NLP	CO3	L2	2.2
2. Explain the working of GANs and diffusion models, and identify their types and use cases in NLP	CO3	L3	2.2
3. Describe the architecture and working of chatbots and dialogue systems using generative techniques	CO3	L2	3.2

Lesson Schedule
Class No. - Portion covered per hour
1. Discriminative vs Generative models
2. Generative Adversarial Networks (GANs) in NLP
3. Types of GANs
4. Diffusion models: concepts & types
5. Introduction to Chatbots and Dialogue System

Review Questions

Sl.No. – Questions	TLO	BL	PI Code
1. Differentiate between discriminative and generative models	TLO1	L3	2.2.4
2. What are Generative Adversarial Networks (GANs)? Explain their working architecture with a diagram.	TLO1	L2	2.2.3
3. List and describe different types of GANs used in NLP tasks.	TLO2	L3	2.2.4
4. What are diffusion models? Explain the core concept and list different types of diffusion models.	TLO2	L2	2.2.3
5. What is a chatbot? Explain the structure of a dialogue system and the difference between retrieval-based and generative chatbots.	TLO3	L3	3.3.2

Course Code and Title: 24EBCC304/Natural Language Processing with GenAI
Chapter Number and Title: Chapter 6. Large Language Models
Planned Hours: 05Hrs
Learning Outcomes: -
At the end of the topic the student should be able to:

TLO's	CO's	BL	CA Code
1. Explain the architecture and functioning of popular LLMs like GPT and BERT	CO3	L2	2.1
2. Apply prompting strategies and fine-tuning techniques such as Adapters and LoRA for customizing LLMs.	CO3	L3	2.2
3. Evaluate the applications, effectiveness, and limitations of LLMs in NLP tasks.	CO3	L4	3.2

Lesson Schedule
Class No. - Portion covered per hour

Introduction to LLMs: GPT

BERT

Prompting techniques for GenAI

Adapters and LoRA (Low-Rank Adaptation)

Applications and limitations of LLMs

Review Questions

Sl.No. – Questions	TLO	BL	PI Code
1.What are prompting techniques in Generative AI? Explain zero- shot, few-shot, and chain-of-thought prompting with examples.	TLO2	L2	2.2.4
2.What are Large Language Models (LLMs)? Explain the architectures of GPT and BERT with key differences.	TLO1	L3	2.1.2
3.Explain Adapters and LoRA (Low-Rank Adaptation) techniques for fine-tuning large language models. How do they differ from full fine-tuning?	TLO2	L2	2.2.4
4.Discuss the key applications and limitations of LLMs in real-world tasks. Provide examples	TLO3	L2	3.2.3

Question Paper Title: Model Question Paper for ISA- I						
Total Duration: 75 Minutes		Course: NLP with GenAI Code: 24EBCC304		Maximum Marks :40		
Note: 1. Answer any two FULL questions 2. Missing data may be suitably assumed with justification.						
Q.No.	Questions	Marks	CO	BL	PO	PI Code
1a	Imagine you are part of a multidisciplinary team building an AI-powered solution for real-world problems such as medical report analysis, financial document summarization, or automated legal contract review. The team has limited understanding of language technologies. as the NLP specialist, define Natural Language Processing and explain how it can be applied to solve challenges in such domains	10	CO1	L2	4	2.1.2
1b	Consider the following corpus of three sentences a. There is a big garden b. Children play in a garden c. They play inside beautiful garden Calculate P for the sentence “They play in a big Garden” assuming bigram language model	10	CO2	L3	2	2.2.3
2a	Draw a basic dependency tree for the sentence “ John eats an apple. ” Explain the dependency relations.	10	CO2	L3	4	2.2.3
2b	You are working on a grammar correction tool for an educational writing app that helps students improve their English sentences. One of the features involves detecting subject–verb agreement errors using Part-of-Speech (POS) tagging. a) What is POS tagging in NLP? Explain its importance.b) Using the sentence: “ The cats is playing in the garden. ”, perform POS tagging .	10	CO1	L2	2	3.1.4
3a	Identify the morphological type (noun phrase, verb phrase, adjective phrase) of following sentence a. Important to Bill b. Looked up the tree	10	CO2	L3	2	2.2.3
3a	Analyze the process of text preprocessing in Natural Language Processing (NLP) with the help of the following sentence: “ The students are studying natural language processing in the lab. ” Describe the steps involved such as tokenization, stop word removal, stemming, and lemmatization.	10	CO1	L3	4	2.2.3

Question Paper Title: Model Question Paper for ISA- II						
Total Duration: 75 Minutes		Course: NLP with GenAI Code: 24EBCC304			Maximum Marks :40	
Note: 1. Answer any two FULL questions 2. Missing data may be suitably assumed with justification.						
Q.N O	Questions	Marks	CO	BL	PO	PI Code
1a	A publishing company wants to build an AI tool that can automatically generate short story endings based on the beginning of the story written by the user. The system must understand and continue the narrative in a coherent way. Explain how Long Short-Term Memory networks can be utilized for such generative writing tasks.	10	CO2	L3	2	2.2.4
1b	Many educational institutions are now turning to conversational AI to streamline student services and reduce manual workload. Explain how such systems are built to handle user queries naturally and effectively. Describe the essential components and flow of a dialogue system that could support this kind of chatbot.	10	CO3	L3	3	3.3.2
2a	What is Machine Translation? Explain its types with real time examples.	10	CO2	L2	4	2.1.2
2b	Explain the architectures of GPT and BERT with key differences	10	CO3	L2	2	2.1.2
3a	List and describe different types of GANs used in NLP tasks.	10	CO3	L3	4	2.2.4
3b	Organizations working with domain-specific language models often face constraints on memory and compute while adapting large pre-trained models.Explain how Low-Rank Adaptation (LoRA) helps in fine-tuning large language models efficiently in such resource-limited environments. Highlight its working principle and advantages over full fine-tuning.	10	CO3	L3	2	2.2.4

Question Paper Title: Model Question Paper for ESA						
Total Duration: 2 Hours		Course: NLP with GenAI Code: 24EBCC304			Maximum Marks :60	
Note: 1. Answer any two FULL questions from each Unit 2. Missing data may be suitably assumed with justification.						
Unit-1						
Q.N O	Questions	Marks	CO	BL	PO	PI Code
1a	Imagine you are part of a multidisciplinary team building an AI-powered solution for real-world problems such as medical report analysis, financial document summarization, or automated legal contract review. The team has limited understanding of language technologies. as the NLP specialist, define Natural Language Processing and explain how it can be applied to solve challenges in such domains	07	CO1	L2	4	2.1.2
1b	Consider the following corpus of three sentences a. There is a big garden b. Children play in a garden c. They play inside beautiful garden Calculate P for the sentence “They play in a big Garden” assuming bigram language model	08	CO2	L3	2	2.2.3
2a	Draw a basic dependency tree for the sentence “ John eats an apple. ” Explain the dependency relations.	07	CO2	L3	4	2.2.3
2b	You are working on a grammar correction tool for an educational writing app that helps students improve their English sentences. One of the features involves detecting subject–verb agreement errors using Part-of-Speech (POS) tagging . a) What is POS tagging in NLP? Explain its importance. b) Using the sentence: “ The cats is playing in the garden. ”, perform POS tagging and identify the grammatical error.	08	CO1	L2	2	3.1.4
3a	Identify the morphological type (noun phrase, verb phase, adjective phrase) of following sentence a. Important to Bill b. Looked up the tree	07	CO2	L3	2	2.2.3
3b	Analyze the process of text preprocessing in Natural Language Processing (NLP) with the help of the following sentence: “ The students are studying natural language processing in the lab. ” Describe the steps involved such as tokenization, stop word removal, stemming, and lemmatization.	08	CO1	L3	4	2.2.3

Unit II						
4a	A publishing company wants to build an AI tool that can automatically generate short story endings based on the beginning of the story written by the user. The system must understand and continue the narrative in a coherent way. Explain how Long Short-Term Memory networks can be utilized for such generative writing tasks.	07	CO2	L3	2	2.2.4
4b	Many educational institutions are now turning to conversational AI to streamline student services and reduce manual workload. Explain how such systems are built to handle user queries naturally and effectively. Describe the essential components and flow of a dialogue system that could support this kind of chatbot.	08	CO3	L3	3	3.3.2
5a	What is Machine Translation? Explain its types with real time examples.	07	CO2	L2	4	2.1.2
5b	Explain the architectures of GPT and BERT with key differences	08	CO3	L2	2	2.1.2
6a	List and describe different types of GANs used in NLP tasks.	07	CO3	L3	4	2.2.4
6b	Organizations working with domain-specific language models often face constraints on memory and compute while adapting large pre-trained models. Explain how Low-Rank Adaptation (LoRA) helps in fine-tuning large language models efficiently in such resource-limited environments. Highlight its working principle and advantages over full fine-tuning.	08	CO3	L3	2	2.2.4